Let $X_1, X_2, \dots, X_n$ be a random sample from a population.

It means that $\mathrm{X}_1, \mathrm{X}_2, \dots, \mathrm{X}_\mathrm{n}$ are independent and identically distributed (iid).

Our next objective is to estimate some unknown parameters of the population by using the statistics, functions of $X_1, X_2, \dots, X_n$.

Before conducting parameter estimation, we need to understand the distributions of common statistical quantities.

A statistic's distribution

If the population distribution is well-defined and the statistic is a simple function of X_i's, then the statistic's distribution can be obtained based on the join distributions of X_i's.

Let $X_1, X_2, \ldots, X_n$ be a random sample from a well-defined distribution. The joint pmf or pdf is the product of the n marginal's since $X_1, X_2, \ldots, X_n$ are independent.

Example (the population distribution has continuous distribution)

Service time for a certain type of bank transaction is a random variable having an exponential distribution with parameter λ. Suppose X_1 and X_2 are service times for two different customers, assumed independent of each other. Consider the total service time $\mathrm{T}_0 = X_1 + X_2$ for the two customers, which is a statistic.

Both X_1 and X_2 have an exponential distribution with parameter λ, and they are independent of each other. So their joint pdf is

$$f(x_1, x_2) = \lambda^2 e^{-\lambda x_1} e^{-\lambda x_2}.$$

And thus the cdf of T_0 is for any $t \geq 0$

$$P(T_0 \leq t) = P(X_1 + X_2 \leq t) = \iint_{x_1 + x_2 \leq t} f(x_1, x_2) dx_1 dx_2$$

$$= \lambda^2 \int_0^t e^{-\lambda x_1} \left(\int_0^{t - x_1} e^{-\lambda x_2} \, dx_2 \right) dx_1$$

$$= 1 - e^{-\lambda t} - \lambda t e^{-\lambda t}$$

And the pdf of T_0 is $f_{T_0}(t) = \begin{cases} \lambda^2 t e^{-\lambda} & t \geq 0 \\ 0 & t < 0 \end{cases}$

Example (the population distribution has discrete distribution)

Suppose the probability distribution of a random rv X is given by

x	80	100	120
$p(x)$	.2	.3	.5

with $\mu = 106$, $\sigma^2 = 244$

Let X_1, X_2 be a random sample from the above distribution. Determine the sampling distribution of $\bar{X}$ and S^2.

Solution Since X_1, X_2 is a random sample, $P(X_1 = x_1, X_2 = x_2) = p(x_1)p(x_2)$

We list the possible values of X_1, X_2 and the probability in the following table.

x_1	x_2	$p(x_1, x_2)$	$\bar{x}$	s^2
80	80	.04	80	0
80	100	.06	90	200
80	120	.10	100	800
100	80	.06	90	200
100	100	.09	100	0
100	120	.15	110	200
120	80	.10	100	800
120	100	.15	110	200
120	120	.25	120	0

And thus, the pmf of $\bar{X}$ is given by

$\bar{x}$	80	90	100	110	120
$p_{\bar{X}}(\bar{x})$	.04	.12	.29	.30	.25

The pmf of S^2 is given by

s^2	0	200	800
$P_{S^2}(s^2)$	0.38	0.42	0.2

Simulation Experiment (not required in the final exam)

The second method of obtaining information about a statistic's distribution is to perform a simulation experiment. This method is usually used when a derivation via probability rules is too difficult or complicated to be carried out.

Such an experiment is usually done with the aid of a computer.

The essence is to obtain k different random samples with size n with appropriate software

For each sample data, calculate the value of the statistic and construct a histogram of the k values. This histogram gives the approximate sampling distribution of the statistic.

5.5 The distribution of a linear combination

The sample mean $\bar{X}$ and sample total T_o are two statistics that are widely used in application.

To understand more about the distribution of $\bar{X}$ and T_0, we will learn the distribution of a linear combination of rv's.

Given a collection of n random variables $X_1, X_2, \ldots, X_n$ and n numerical constants $a_1, a_2, .., a_n$, the rv

$$Y = a_1 X_1 + \cdots + a_n X_n$$

is called a linear combination of the $X_i's$.

Suppose $X_1, X_2, \dots, X_n$ have the expected values $\mu_1, \mu_2, \dots, \mu_n$ respectively, and variances $\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2$, respectively.

1. Whether or not the X_i's are independent

$$E(a_1X_1 + \cdots + a_nX_n) = a_1E(X_1) + \cdots + a_nE(X_n)$$

$$= a_1\mu_1 + \cdots + a_n\mu_n.$$

Proofs for the Case $n = 2$ and X_1, X_2 are both continuous rv.

$$E(a_1X_1 + a_2X_2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(a_1x_1 + a_2x_2)\cdot f(x_1,x_2)dx_1dx_2$$

$$= a_1\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}x_1\cdot f(x_1,x_2)dx_1dx_2 + a_2\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}x_2\cdot f(x_1,x_2)dx_1dx_2$$

$$= a_1E(X_1) + a_2E(X_2).$$

2. If X_i's are independent (or just uncorrelated)

$$V(a_1X_1 + \cdots + a_nX_n) = a_1^2V(X_1) + \cdots + a_n^2V(X_n)$$
$$= a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2$$

Proofs for the Case $n = 2$.

$$V(a_1X_1 + a_2X_2) = E\left[\left(a_1X_1 + a_2X_2 - \mu_{a_1X_1+a_2X_2}\right)^2\right]$$

$$= E\left[\left(a_1(X_1 - \mu_{X_1}) + a_2(X_2 - \mu_{X_2})\right)^2\right]$$

$$= a_1^2E\left[(X_1 - \mu_{X_1})^2\right] + 2a_1a_2E[(X_1 - \mu_{X_1})(X_2 - \mu_{X_2})] + a_2^2E\left[(X_2 - \mu_{X_2})^2\right]$$

$$= a_1^2V(X_1) + a_2^2V(X_2) + 2a_1a_2Cov(X_1, X_2)$$

$$= a_1^2V(X_1) + a_2^2V(X_2)$$

3. For any $X_1, X_2, \dots, X_n$

$$V(a_1X_1 + \cdots + a_nX_n) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j Cov(X_i, X_j)$$

A special case: the Difference Between Two Random Variables

Let $Y = X_1 - X_2$. We then have the following result.

$E(Y) = E(X_1) - E(X_2)$ for any two rv's X_1 and X_2.

$V(Y) = V(X_1) + V(X_2)$ if X_1 and X_2 are two independent rv's.

Example A gas station sells three grades of gasoline: regular, extra, and super. These are priced at $\$3.00, \$3.20,$ and $\$3.40$ per gallon, respectively. Let X_1, X_2 and X_3 denote the amounts of these grades purchased (gallons) on a particular day. Suppose the X_i's are independent with $\mu_1 = 1000$, $\mu_2 = 500$, $\mu_3 = 300$, $\sigma_1 = 100$, $\sigma_2 = 80$ and $\sigma_3 = 50$. The revenue from sales is $Y = 3X_1 + 3.2X_2 + 3.4X_3$. Then

$$E(Y) = 3\mu_1 + 3.2\mu_2 + 3.4\mu_3 = \$5620$$

$$V(Y) = (3)^2\sigma_1^2 + (3.2)^2\sigma_2^2 + (3.4)^2\sigma_3^2 = 184{,}436$$

$$\sigma_Y = \sqrt{V(Y)} = \$429.46$$

The Distribution of the linear combination of independent normals

If $X_1, X_2, \dots, X_n$ are independent, normally distributed rv's (with possibly different means and/or variances), then any linear combination of the X_i's also has a normal distribution. In particular, the difference $X_1 - X_2$ between two independent, normally distributed variables is normally distributed.

Proofs for the Case $Y = X_1 - X_2$ where X_1, X_2 are independent, standard normally distributed rv's.

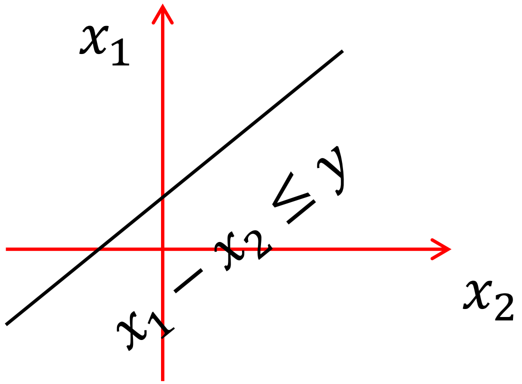
The joint pdf of X_1 and X_2 is

$$f(x_1, x_2) = \frac{1}{2\pi} e^{-x_1^2/2} e^{-x_2^2/2}$$

For any y

$$P(Y \le y) = \iint_{x_1 - x_2 \le y} f(x_1, x_2) dx_1 dx_2$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{y+x_2} e^{-x_2^2/2} e^{-x_1^2/2}\, dx_1 dx_2$$



The pdf of Y is given by

$$f_Y(y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-x_2^2/2}\, e^{-(y+x_2)^2/2} \mathrm{dx}_2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(2x_2^2 + 2yx_2 + y^2)/2}\, dx_2$$

$$= \frac{1}{\sqrt{4\pi}} e^{-y^2/4} \cdot \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(x_2 + y/2)^2}\, dx_2 \qquad = \frac{1}{\sqrt{4\pi}} e^{-y^2/4}$$

Obviously, $Y = X_1 - X_2 \sim N(0,2)$.

The more general cases can be proved in a similar way.

Example

If X and Y are two independent rv's, $X \sim N(2{,}0.2)$ and $Y \sim N(2{,}0.2)$, then what is the pdf of the rv $Z = X - 2Y + 1$?

Solution

$$E(Z) = E(X) - 2E(Y) + 1 = -1$$

$$V(Z) = V(X) + 4V(Y) = 0.2 + 4(0.2) = 1$$

And thus the $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-(z+1)^2/2}$

The Distribution of the Sample Mean

In the next chapter, we will use the sample mean $\bar{X}$ to estimate the population mean μ.

Before the data becomes available, the sample observations $X_1, X_2, \dots, X_n$ are rv's. And thus the sample mean $\bar{X}$ is also a random variable.

Observe that $\bar{X} = (X_1 + X_2 + \cdots + X_n) \cdot \frac{1}{n}$. We can easily obtain the following proposition, which gives the relationships between $E(\bar{X})$ and μ, and also among $V(\bar{X})$, σ^2 and n.

proposition Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean value μ and standard deviation σ. Then

1. $E(\bar{X}) = \mu_{\bar{X}} = \mu$
2. $V(\bar{X}) = \sigma_{\bar{X}}^2 = \sigma^2/n$ and $\sigma_{\bar{X}} = \sigma/\sqrt{n}$

According to Result 1, the distribution of $\bar{X}$ is centered precisely at the mean of the population from which the sample has been selected.

Result 2 shows that the distribution becomes more concentrated about μ as the sample size n increases.

The standard deviation $\sigma_{\bar{X}}$ is often called the standard error of the mean.

Let $T_0 = X_1 + X_2 + \cdots + X_n$ (the sample total). $T_0 = n\bar{X}$, and then

$E(T_0) = n\mu$ and $\sigma_{T_0} = \sqrt{n}\sigma$.

The distribution of T_0 becomes more spread out as n increases.

The Case of Normal Distributions

Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and standard deviation σ. Then for any n, $\bar{X}$ is normally distributed (with mean μ and standard deviation $\sigma/\sqrt{n}$), as is T_o (with mean $n\mu$ and standard deviation $\sqrt{n}\sigma$).

Since both $\bar{X}$ and T_o are linear combination of independent, normally distributed rv's $X_1, X_2, \ldots, X_n$, both $\bar{X}$ and T_o have a normal distribution.

Observe that $E(\bar{X}) = \mu$, $V(\bar{X}) = \sigma^2/n$, $E(T_0) = n\mu$ and $\sigma_{T_0} = \sqrt{n}\sigma$. We have

$\bar{X} \sim N(\mu, \sigma^2/n)$ and $T_o \sim N(n\mu, n\sigma^2)$.

Example

The time that it takes a randomly selected rat of a certain subspecies to find its way through a maze is a normally distributed rv with $\mu = 1.5$ min and $\sigma = 0.35$ min. Suppose five rats are selected. Let $X_1, X_2, \ldots, X_5$ denote their times in the maze. What is the probability that the total time $T_0 = X_1 + \cdots + X_5$ for the five is between 6 and 8 min?

$T_o \sim N(n\mu, n\sigma^2) = \mathrm{N}(7.5, 0.6125)$. Then

$$P(6 \le T_0 \le 8) = P\left(\frac{6-7.5}{\sqrt{0.6125}} \le \frac{T_0-7.5}{\sqrt{0.6125}} \le \frac{8-7.5}{\sqrt{0.6125}}\right)$$

$$= P(-1.92 \le Z \le 0.64)$$

$$= \Phi(0.64) - \Phi(-1.92) = 0.7115$$

Determine the probability that the sample average time $\bar{X}$ is at most 2.0 min.

$\bar{X} \sim N(\mu, \sigma^2/5) = N(1.5, 0.0245)$. Then

$$P(\bar{X} \leq 2) = P\left(\frac{\bar{X}-1.5}{\sqrt{0.0245}} \leq \frac{2-1.5}{\sqrt{0.0245}}\right)$$

$$= P(Z \leq 3.19)$$

$$= \Phi(3.19) = 0.9993$$

The Central Limit Theorem

When the X_i's are independently and normally distributed, so is $\bar{X}$ for every sample size n.

In application, we are usually interested to know the distribution of $\bar{X}$ if the sample is not from a normal distribution.

In Chapter 3, we know that if $X_i's$ are iid and each X_i is a Bernoulli rv with $P(X_i = 1) = p$, then $T_0 = X_1 + \cdots + X_n \sim Bin(n, p)$.

In Chapter 4, we compare the probability histogram of T_0 with $n = 20$ and $p = 0.6$ and a normal curve with $\mu = np = 12$ and $\sigma = \sqrt{np(1-p)} = 2.19$.

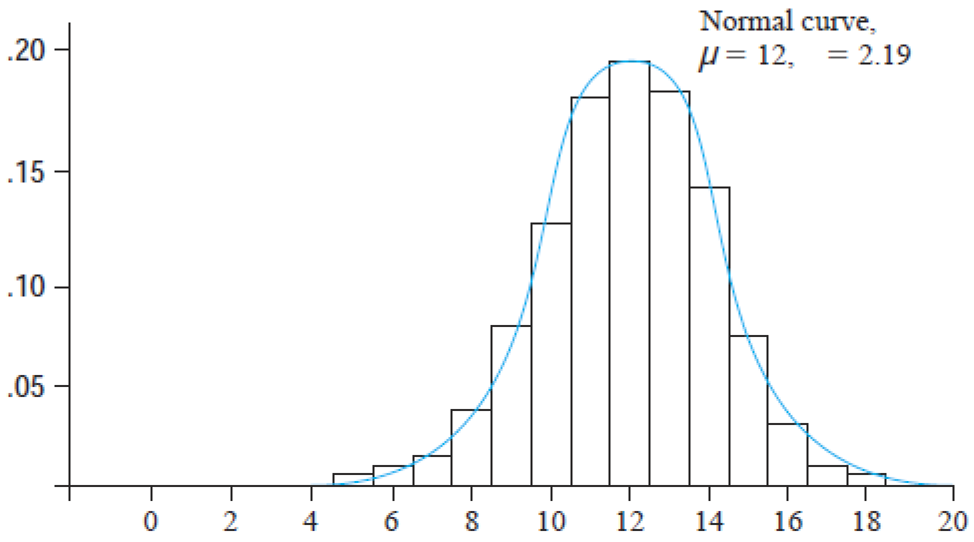


Figure 4.25 Binomial probability histogram for $n = 20$, $p = .6$ with normal approximation curve superimposed

The figure suggests that even when the population distribution is not normal, $T_0 = X_1 + \cdots + X_{20}$ has approximately a normal distribution.

Since $\bar{X} = \frac{1}{n}T_0$, $\bar{X}$ also has approximately a normal distribution.

The formal statement of this result is the most important theorem of probability.

The Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Then if n is sufficiently large, $\bar{X}$ has approximately a normal distribution with $\mu_{\bar{X}} = \mu$ and $\sigma_{\bar{X}}^2 = \sigma^2/n$, and T_0 also has approximately a normal distribution with $\mu_{T_0} = n\mu$ and $\sigma_{T_0} = \sqrt{n}\sigma$. The larger the value of n, the better the approximation.

补充：由CLT可直观感知，当样本容量足够大时，样本均值收敛于总体期望，这就是大数定律。大数定律告诉我们，可以用样本均值近似代替总体期望，可以用随机事件在足够多次试验中出现的频率近似代替概率。

CLT还告诉我们，用样本均值近似代替总体期望的标准误差。

A practical difficulty in applying the CLT is in knowing when n is sufficiently large.

If the underlying distribution is close to a normal density curve, then the approximation will be good even for a small n, whereas if it is far from being normal, then a large n will be required.

$X_i \sim Bin(1, p)$. When p is close to 0.5, the distribution of each X_i is reasonably symmetric, the approximation is good if $0.5n > 10$, i.e., $n > 20$. When p is 0.01, the distribution of each X_i is quite skewed, the approximation is good if $0.01n > 10$ and $(1 - 0.01)n > 10$, i.e., $n > 100$.

Example The net weight of each bag of salt is a random variable, with an average weight of 100g and a standard deviation of 10g. What is the probability that the net weight of 200 bags of salt is bigger than 20500g?

Let X_i be the net weight of ith bag of salt. X_i 's are independent and identically distributed rv's.

$$E(X_i) = 100 \text{ and } V(X_i) = 100$$

By CLT, the desirable probability is

$$P\left(\sum_{i=1}^{200} X_i > 20500\right) = 1 - P\left(\sum_{i=1}^{200} X_i \le 20500\right)$$

$$= 1 - P\left(\frac{1}{10\sqrt{200}}\left(\sum_{i=1}^{200} X_i - 200\times 100\right) \le \frac{1}{10\sqrt{200}}(20500 - 200\times 100)\right)$$

$$\approx 1 - \Phi\left(\frac{20500 - 200\times 100}{10\sqrt{200}}\right) = 0.0002$$

Recall that X has a lognormal distribution if $\ln(X)$ has a normal distribution.

Proposition

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution for which only positive values are possible ($P(X_i > 0) = 1$). Then if n is sufficiently large, the product $Y = X_1 X_2 \ldots X_n$ has approximately a lognormal distribution.

Observe that $\ln(Y) = \ln(X_1) + \cdots + \ln(X_n)$

Since $\ln(Y)$ is a sum of independent and identically distributed rv's. It is approximately normal when n is large, so Y itself has approximately a lognormal distribution.

Chapter 6 The point estimation (点估计)

Suppose we want to estimate a parameter θ of a single population.

One possible way is to select a suitable point estimator $\hat{\theta}$ and to compute its value from the given sample data, which can be regarded as a sensible value for θ.

In general, we have two “constructive” methods for obtaining point estimators: the method of moments (矩法估计) and the method of maximum likelihood (极大似然估计).

The Method of Moments (矩法估计)

Let $X_1, X_2, \ldots, X_n$ be a random sample from a pmf or pdf $f(x)$. For $k = 1, 2, 3, \ldots$, the kth population moment (k阶原点矩), or kth moment of the distribution, is $E(X^k)$. The kth sample moment is $(1/n)\sum_{i=1}^{n} X_i^k$.

The population moments will be functions of any unknown parameters $\theta_1, \theta_2, \ldots$

The basic idea of this method is to equate the kth sample moment , to the kth population moment. Then solve these equations to obtain the estimators.

For example, we want to estimate the population moment μ. We can equate the first sample moment $\bar{X} = \sum X_i / n$ to μ. The resulting equation is $\mu = \bar{X}$, which gives an estimator of μ.

Example (only one parameter to be estimated)
Let $X_1, X_2, \ldots, X_n$ represent a random sample of service times of n customers at a certain facility, where the underlying distribution is exponential with parameter λ.

Since there is only one parameter to be estimated, the estimator can be obtained by equating $E(X)$ to $\bar{X}$.

Observe that $E(X) = 1/\lambda$ for an exponential distribution. This gives $1/\lambda = \bar{X}$. Solving this equation gives $\lambda = 1/\bar{X}$. Then we obtain the moment estimator (点估计量) of λ: $\hat{\lambda} = 1/\bar{X}$.

If a sample data becomes available, we can compute the value of the estimator of λ, that represents a sensible value for λ and is called a point estimate (点估计值) of λ.

Suppose the observed service times (min) of 3 customers were $x_1 = 5$, $x_2 = 6$, $x_3 = 7$.

Use of the moment estimator $\hat{\lambda} = 1/\bar{X}$ would have resulted in an estimate $\hat{\lambda} = \frac{3}{5+6+7} = \frac{1}{6}$.

Notation

The symbol $\hat{\lambda}$ is used to denote both the estimator of λ and the point estimate resulting from a given sample.

Thus $\hat{\lambda} = 1/\bar{X}$ is read as “the point estimator of λ is the inverse of the sample mean $\bar{X}$”.

$\hat{\lambda} = \frac{1}{6}$ means the point estimate of λ is $\frac{1}{6}$

Another example from the final exam

VII. (10 points) Let $X_1, X_2, \cdots, X_n$ be a random sample of size n from the pdf

$$f(x;\theta) = \begin{cases} \frac{x}{\theta} e^{-\frac{x^2}{2\theta}}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) Use the method of moments to find an estimator for θ.

(b) Find the maximum likelihood estimator for θ.

$$E(X) = \int_{-\infty}^{+\infty} x f(x,\theta) dx = \int_0^{+\infty} \frac{x^2}{\theta} e^{-\frac{x^2}{2\theta}} dx = \frac{1}{2}\int_{-\infty}^{+\infty} \frac{x^2}{\theta} e^{-\frac{x^2}{2\theta}} dx$$

$$= \frac{\sqrt{2\pi}}{2\sqrt{\theta}} \int_{-\infty}^{+\infty} \frac{x^2}{\sqrt{2\pi}\sqrt{\theta}} e^{-\frac{x^2}{2\theta}} dx = \sqrt{\frac{\pi}{2\theta}} \cdot V(Y),\ (Y \sim N(0,\theta))$$

$$= \sqrt{\frac{\pi}{2\theta}} \cdot (\sqrt{\theta})^2 = \sqrt{\frac{\theta \cdot \pi}{2}}$$

$$\therefore \theta = \frac{2E^2(x)}{\pi}.$$

Thus, the moment estimator is $\hat{\theta}_M = \frac{2}{\pi}\overline{X}^2$.

方法2：$E(X)$求不出来，可以试试求$E(X^2)$

(or using 2rd moment)

$$
\begin{aligned}
E(X^2) &= \int_{-\infty}^{+\infty} x^2 f(x,\theta)dx = \int_0^{+\infty} \frac{x^3}{\theta} e^{-\frac{x^2}{2\theta}} dx \\
&= \int_0^{+\infty} x^2 e^{-\frac{x^2}{2\theta}} d\frac{x^2}{2\theta} = 2\theta \cdot \int_0^{+\infty} \frac{x^2}{2\theta} e^{-\frac{x^2}{2\theta}} d(\frac{x^2}{2\theta}) \\
&= 2\theta \cdot \int_0^{+\infty} ue^{-u} du = 2\theta
\end{aligned}
$$

$$\therefore \theta = \frac{E(X^2)}{2}$$

The moment estimator is $\hat{\theta}_M = \frac{1}{2n}\sum_{i=1}^n X_i^2$.

More general case

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pmf or pdf $f(x; \theta_1, \ldots, \theta_m)$, where $\theta_1, \ldots, \theta_m$ are parameters whose values are unknown. Then the moment estimators $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are obtained by equating the first m sample moments to the corresponding first m population moments and solving for $\theta_1, \ldots, \theta_m$.

补充：在实际应用中，我们也常常用样本方差估计总体方差。总体方差称为总体中心矩。

Example (There are more than one parameters to be estimated)
Let $X_1, X_2, \ldots, X_n$ be a random sample from a negative binomial distribution with parameters r and p. State the moment estimators of r and p (both r and p are unknown).

Observe that $E(X) = \frac{r}{p}$, $\quad V(X) = r(1-p)/p^2$
and thus $E(X^2) = V(X) + [E(X)]^2 = r(1-p+r)/p^2$

Equating $E(X) = \bar{X}$ and $E(X^2) = 1/n \sum X_i^2$, we obtain two equations with unknowns r and p. Solving these two equations yeilds the point estimators of r and p:

$$\hat{p} = \frac{\bar{X}}{\left(\frac{1}{n}\right)\sum X_i^2 - \bar{X}^2 + \bar{X}} \qquad \hat{r} = \frac{\bar{X}^2}{\left(\frac{1}{n}\right)\sum X_i^2 - \bar{X}^2 + \bar{X}}$$

r means the number of successes, and thus must be nonnegative. However, the moment estimators $\hat{r}$ may result in negative estimate of r. This indicates that the moment estimator is flawed.